

NRS for every conjugate pair

The Robertson–Schrödinger table on $T_d : P_d$ at maximal tension

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Lean 4 package nava-robertson-schrodinger, module D39_ConjugatePairs

1 What the table says

Robertson (1929) and Schrödinger (1930) proved, for any two observables A, B and any normalized state ψ ,

$$\sigma_A^2 \sigma_B^2 \geq \left(\frac{1}{2} \langle \{A, B\} \rangle - \langle A \rangle \langle B \rangle \right)^2 + \left(\frac{1}{2} |\langle [A, B] \rangle| \right)^2 = |\langle \tilde{A} \psi, \tilde{B} \psi \rangle|^2,$$

where $\tilde{A}\psi = (A - \langle A \rangle)\psi$ and $\tilde{B}\psi = (B - \langle B \rangle)\psi$ are the fluctuation vectors. The theorem carries no list of pairs: it holds for every pair (D2_Robertson). NRS evaluates it on the elemental pair $T_d : P_d$ at the maximal current state, and measures the ratio

$$R = \frac{\sigma_A \sigma_B}{|\langle \tilde{A} \text{maxCurrentState}, \tilde{B} \text{maxCurrentState} \rangle|} \geq 1.$$

$R = 1$ means the inequality is saturated; $R > 1$ is what the inequality cannot close.

The result of this note. R is the same number $C_{\text{Nava}}(d)$ for *every* conjugate pair realized on $T_d : P_d$. It equals 1 exactly at $d = 2, 3$; from $d = 4$ on it is strictly larger than 1, grows strictly with d , and stays strictly below $C_\infty = \sqrt{\pi^2/3 - 2}$, which no d reaches. On a cube of three pairs the leftover is the volumetric quantum $\mathcal{V}$: zero at $3 \times 3 \times 3$, strictly positive from $4 \times 4 \times 4$, growing with every axis, never vanishing. All of this is proved in Lean 4 with only the standard axioms.

2 The setting

On $H_d = \mathbb{C}^d$:

- P_d is diagonal with the d equispaced values of $[-1, 1]$: it orders the values of the pair.
- $T_d = A_d/\rho_d$ is the adjacency of the path on d sites, $\rho_d = 2 \cos(\pi/(d+1))$: the step between consecutive values. It is the only graph compatible with ordered locality and completeness (D3_PathGraph).
- $\text{maxCurrentState}_j \propto (-i)^j \sin((j+1)\pi/(d+1))$ is the maximal current state (D5_MaximalTension, D21); at it, $\frac{1}{2} |\langle [T_d, P_d] \rangle| = 1/(d-1)$.

A pair realized on $T_d : P_d$ is

$$A = a T_d + b, \quad B = c P_d + e, \quad a, c \in \mathbb{R} \setminus \{0\}, \quad b, e \in \mathbb{R}.$$

The numbers a, c are the units of the pair and b, e its origins: metres and kg m/s for position–momentum, a count and radians for number–phase, coulombs and webers for charge–flux, and so on.

3 The theorem

Theorem (D39_ConjugatePairs). *Let $d \geq 2$ and $a, c \neq 0$. For the pair $A = aT_d + b$, $B = cP_d + e$ at ψ^* :*

1. $R = C_{\text{Nava}}(d)$ (ratio_pair);
2. *the fluctuation vectors are parallel ($R = 1$) if and only if $d = 2$ or $d = 3$* (saturated_pair_iff);
3. *for $d \geq 4$ they meet at an angle $\theta_{\text{NRS}}(d) = \arccos(1/C_{\text{Nava}}(d)) > 0$* (angle_pair_pos);
4. *the angle grows strictly with d , even if the pair at d and the pair at d' carry different units* (angle_pair_lt_of_lt);
5. *it stays strictly below $\arccos(1/C_\infty) \approx 28.30^\circ$, which no d attains* (angle_pair_lt_limit);
6. *the excess $R - 1$ is the dimensional quantum $\delta(d) = C_{\text{Nava}}(d) - 1$* (dimQuantum_pair).

On the cube $d_x \times d_y \times d_z$ with one pair per axis, $A_i = a_i T_{d_i} + b_i$, $B_i = c_i P_{d_i} + e_i$ (all $a_i, c_i \neq 0$, each axis in its own units) and the maximal current state of the cube, the product of the maximal current states of the axes:

7. *axis i has ratio $C_{\text{Nava}}(d_i)$* (ratio_pair_axis_x/y/z);
8. $(R_x - 1)(R_y - 1)(R_z - 1) = \mathcal{V}(d_x, d_y, d_z) = \delta(d_x) \delta(d_y) \delta(d_z)$ (volQuantum_pairs);
9. $\mathcal{V} = 0$ *if and only if some axis has 2 or 3 sites; from $4 \times 4 \times 4$ on, $0 < \mathcal{V} < \delta_\infty^3$* (volQuantum_pairs_certificate).

Why units do not matter. For a unit state, $\langle aL + b \rangle = a\langle L \rangle + b$, so the fluctuation vector of $aL + b$ is exactly a times that of L (centeredG_affineOp). In $R = \|\tilde{A}\psi\| \|\tilde{B}\psi\| / |\langle \tilde{A}\psi, \tilde{B}\psi \rangle|$ the factor $|a| |c|$ appears once in the numerator and once in the denominator and cancels (ratioG_affineOp); the same holds for the angle (angleG_affineOp). These two lemmas hold at *every* unit state, not only at the maximal current state, so any statement about $T_d : P_d$ that depends only on R or on the angle, such as strictness on the band of states near maximal tension (D23d), holds for every pair as well.

The only thing a pair changes is its floor,

$$\frac{1}{2} |\langle [A, B] \rangle| = |ac| \cdot \frac{1}{2} |\langle [T_d, P_d] \rangle| = \frac{|ac|}{d-1},$$

expressed in the units of that pair.

4 The table

d	$R = C_{\text{Nava}}(d)$	excess $\delta(d)$	angle $\theta_{\text{NRS}}(d)$	cube $\mathcal{V}(d, d, d) = \delta(d)^3$	floor of a pair
2	1	0 (saturates)	0°	0	$ ac $
3	1	0 (saturates)	0°	0	$ ac /2$
4	1.008479	0.85 %	7.43°	6.09×10^{-7}	$ ac /3$
5	1.018350	1.84 %	10.89°	6.18×10^{-6}	$ ac /4$
6	1.027727	2.77 %	13.34°	2.13×10^{-5}	$ ac /5$
8	1.043563	4.36 %	16.61°	8.27×10^{-5}	$ ac /7$
10	1.055806	5.58 %	18.71°	1.74×10^{-4}	$ ac /9$
20	1.088392	8.84 %	23.25°	6.91×10^{-4}	$ ac /19$
50	1.114641	11.46 %	26.21°	1.51×10^{-3}	$ ac /49$
100	1.124785	12.48 %	27.24°	1.94×10^{-3}	$ ac /99$
ceiling	$\sqrt{\pi^2/3} - 2 \approx 1.135724$	13.57 %	28.30°	$\delta_\infty^3 \approx 2.50 \times 10^{-3}$	—

How to read each column.

- $R = C_{\text{Nava}}(d)$: the measured ratio $\sigma_A \sigma_B / |\langle \tilde{A} \text{maxCurrentState}, \tilde{B} \text{maxCurrentState} \rangle|$. It is the same for every row of the catalogue in Section 5.
- Excess $\delta(d)$: how far above the Robertson–Schrödinger floor the pair stays. Zero only at $d = 2, 3$; from $d = 4$ on it is an algebraic quantum that no choice of units removes (D25).
- Angle $\theta_{\text{NRS}}(d)$: the opening between the fluctuation vectors of A and B . At $d = 2, 3$ they are parallel. From $d = 4$ they open, at least 7.43° , and the opening grows with d (D37b).
- $\mathcal{V}(d, d, d)$: the volumetric quantum of the cube with three pairs, one per axis (D37e). It is the box $\delta \times \delta \times \delta$. In the pictures of the repository the six fluctuation vectors of the three axes form a star whose envelope is an octahedron; each of its three axes opens by $\theta_{\text{NRS}}(d)$.
- Floor of a pair: $\frac{1}{2} |\langle [A, B] \rangle|$ in the units of that pair.

The leftover does not disappear. At $3 \times 3 \times 3$ nothing is left ($\mathcal{V} = 0$, all three angles 0°). From $4 \times 4 \times 4$ something is always left: $\mathcal{V} \geq \delta(4)^3$ (`volQuantum_floor`) and every axis opens at least 7.43° (`angle_floor_cube`). Adding sites never shrinks it: $\mathcal{V}$ grows strictly in each axis (`volQuantum_strictMono_x/y/z`) and the angle grows strictly with d (`angleNRS_strictMono0n`). It rises towards the ceiling $\delta_\infty^3 \approx 2.50 \times 10^{-3}$ and 28.30° , and no cube reaches it (`volQuantum_lt_ceiling`, `angleNRS_lt_limit`). The ceiling is a bound approached by finite lattices, not a lattice of its own.

5 The catalogue of pairs

Each pair below is realized on $T_d: P_d$ with its own units a, c and origins b, e ; by the theorem, all rows share the table of Section 4.

#	Pair	P_d : the ordered values	T_d : the step
<i>Position and its equivalents</i>			
1	position x – momentum p_x	cell along x	hop to the next cell
2	position y – momentum p_y	cell along y	hop to the next cell
3	position z – momentum p_z	cell along z	hop to the next cell
4	centre of mass – total momentum; relative coordinate – relative momentum	cell of the coordinate	hop
5	lattice site – lattice momentum (tight-binding chain, open ends)	site	hop
6	waveguide index – transverse momentum	guide	evanescent coupling
7	site of a spin chain, one-excitation sector – exchange	excited site	exchange with the neighbour
8	level n of a d -level system – drive between consecutive levels	level	drive $n \leftrightarrow n+1$
<i>Fields</i>			
9	field $\varphi(x)$ – conjugate momentum $\pi(x)$, site by site	field value	step of field value
10	vector potential A – electric field E , mode by mode	value of A	step of A
11	quadrature X – quadrature Y of a mode	value of X	step of X
12	photon number n – quadrature	photon number	$n \leftrightarrow n+1$

#	Pair	P_d : the ordered values	T_d : the step
13	link electric field E – link variable U (lattice gauge theory)	value of E	$U: E \leftrightarrow E+1$
<i>Number and phase</i>			
14	photon number n – phase φ	photon number	phase shift $n \leftrightarrow n+1$
15	Cooper-pair number n – Josephson phase φ	charge state	tunnelling $n \leftrightarrow n+1$
16	charge Q – flux Φ (LC circuit)	value of Q	step of Q
17	number difference – relative phase (two-mode condensate)	number difference	one particle between modes
<i>Angle, rotation and spin</i>			
18	angle φ – angular momentum L_z	value m of L_z	$m \leftrightarrow m+1$
19	angle – action (action–angle variables)	value of the action	step of the action
20	clock Z – shift X (Weyl pair, d levels; D12)	clock value	shift by one
21	$\sigma_z - \sigma_x$ of a qubit	± 1	flip; exactly (P_2, T_2)
22	$J_z - J_x$ (and cyclic), spin or angular momentum	value m of J_z	$m \leftrightarrow m+1$
23	isospin and colour generators	weight along one generator	step between weights
<i>Magnetic field</i>			
24	kinetic momenta $\pi_x - \pi_y$ (Landau levels)	Landau level	$n \leftrightarrow n+1$
25	guiding centre $X - Y$	guiding-centre level	$n \leftrightarrow n+1$
<i>Time: the dynamics on the axes, not a fourth axis</i>			
26	time t – energy E	t : parameter of $U(t) = e^{-itT_d}$ (D37g)	energy: T_d itself, spectrum $\varepsilon(\theta)$ (D38)
27	time – frequency (Fourier, Gabor)	t , as in row 26	frequency $\varepsilon(\theta_k)$ of each mode of T_d

Space and time. The three axes d_x, d_y, d_z are space: the Hilbert space $\mathbb{C}^{d_x} \otimes \mathbb{C}^{d_y} \otimes \mathbb{C}^{d_z}$, discretized in d points per axis, is the three-dimensional path graph of D37. Rows 1–25 are pairs placed on an axis. Time is not a fourth axis: it is the parameter of the evolution generated by transport, as in the theorem of 1929–1930, where time is not an operator. Counted in steps of transport it gives the light cone (lightCone, lightCone_cube, D37f); in continuous time $U(t) = e^{-itT_d}$ obeys the Lieb–Robinson bound (lieb_robinson, D37g). Energy is T_d : its sine modes are its eigenvectors with energy $\varepsilon(\theta) = 2 \cos \theta / \rho_d$, the dispersion relation, which is also the frequency of each mode (Td_mulVec_sineMode, D38). Time and transport are tied by the Heisenberg equation $\frac{d}{dt}(U^\dagger P_d U) = U^\dagger K_d U$ with $K_d = i[T_d, P_d]$ (heisenberg): the tension is velocity, no state moves faster than one site per unit of time, and the maximal current state moves at exactly that speed (abs_velocity_le, velocity_maxCurrentState); on the cube the maximal current state does so on all three axes at once (velocities_maxCurrentCubeState). So rows 26 and 27 are the dynamics that T_d generates on the axes where rows 1–25 live.

Rows 1–3 are the three axes of the cube of D37. Any three rows among 1–25 whose pairs commute with each other can occupy the three axes, and items 7–9 of the theorem apply to them.

6 Reproduce

```
lake exe cache get
```

```
lake build
```

```
# the whole package
```

```
lake env lean Verification/Layer1_Mathematics.lean # boundary + axioms
```

Every theorem cited here depends only on `propext`, `Classical.choice` and `Quot.sound`; the package has no sorry. The numbers of the table are $C_{\text{Nava}}(d)$ from D9/D21, recomputed from the matrices T_d , P_d and

the maximal current state; the exact value at $d = 4$ is $C_{\text{Nava}}(4) = \sqrt{(99 - 42\sqrt{5})/5}$ (D37b, D37e).